Name: ____________________

Date: ____________________

Mr. Rawson • Y8 MYP Physics

1.1 – Light Sources & Reflectors

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can distinguish between **light sources** and **reflectors**.
- b) I can explain why objects are visible only when light enters my eyes.
- c) I can use the model of a **light ray** to describe how light travels in straight lines.

Theory

Light is a form of energy that allows us to see. However, we do not see objects because they emit light; we see them because light enters our eyes. To understand this, we must categorise objects into two groups: **light sources** and **reflectors**. A **light source** is an object that produces its own light through processes such as combustion, electricity, or nuclear fusion. Common examples include the Sun, a lit filament lamp, and a candle flame. When you switch on a torch, it becomes a light source.

In contrast, a **reflector** does not produce any light of its own. Instead, it bounces (reflects) light that falls upon it from another source. The Moon is the most famous example of a reflector; it shines because it reflects sunlight. Other everyday examples include mirrors, white walls, and your own hand. A crucial piece of evidence for this distinction is that you cannot see a reflector in complete darkness. If you place a mirror in a pitch-black room with no light sources, the mirror remains invisible because there is no light for it to bounce back to your eyes.

Light travels from its source to our eyes (or to a reflector) in a very specific way. Evidence suggests that light travels in **straight lines**. We can observe this through the sharp edges of shadows cast by opaque objects. If light could bend around corners easily, shadows would have fuzzy, blurred edges. Instead, we see distinct dark shapes. Similarly, when you shine a laser pointer or use a ray box in a dusty room, the beam appears as a perfectly straight line. You also cannot see round corners because light does not curve to follow the wall; it travels straight until it hits an object or enters your eye.

To help us model and predict how light behaves, scientists use the concept of a **light ray**. A light ray is a simplified representation used in diagrams to show the path that light takes. We draw light rays as straight lines with arrows indicating the direction of travel. This model helps us explain phenomena like shadows and reflections without needing to track every single photon of light. By using these straight-line models, we can accurately predict where shadows will form and how images will appear in mirrors.

Word bank: light source · reflector · straight lines · light ray · darkness

Activity

Investigating Light Paths with a Ray Box

You will need: a ray box (or laser pointer), a piece of card with a narrow slit cut into it, and a sheet of white paper.

- Place the white paper on your desk in a dimly lit room.
- Position the ray box so that it shines through the slit onto the paper. You should see a single, thin beam of light appearing on the paper.
- Observe the shape of the beam. Is it curved or straight? Trace the path of the light using a pencil to create a **light ray** diagram on your paper.

- Place an opaque object (like a book) in the path of the beam. Observe the shadow formed behind the object. Note how sharp the edge of the shadow is compared to the soft edge you might see from a large, diffuse light source like a window on a cloudy day.
- Move the ray box closer to the slit and then further away. Does the width of the beam change? Does it remain straight?

Observation	**What did you see?**
Shape of the beam	____________
Edge of the shadow	____________
Effect of moving the source	____________

Questions

1. Classify the following objects as either a **light source** or a **reflector**: the Sun, a polished spoon, a firework exploding, a white piece of paper, a computer screen that is turned off.

2. Explain why you can see your friend in a brightly lit room but not in a completely dark room, even though your friend has not moved.

3. If light travelled in curved lines like sound waves around a corner, what would happen to the shadows cast by a tree on a sunny day? Describe the difference.

4. Why is it useful for scientists to use the model of a **light ray** (a straight line with an arrow) instead of drawing every individual particle of light?

Extension

Find out why the sky appears blue during the day but red or orange during sunrise and sunset, considering how light travels through the Earth's atmosphere.